

3

In this chapter, we will learn about

- some properties of materials in nano scale
- applications and potential hazards in nanotechnology
- the working principle of a transmission electron microscope and a scanning tunnelling microscope

Probing into Nano Scale

3.1 Nano objects and nanotechnology

Video

→ This video demonstrates the Hands-on activity.

Hands-on activity

Honey on lotus leaf

What you need

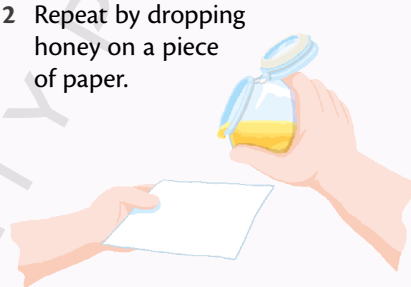


What to do

1 Drop some honey on the lotus leaf and tilt the leaf.



2 Repeat by dropping honey on a piece of paper.



What you find

What happens to the honey on the leaf and on the paper?

Flipped classroom

Watch a video and answer the questions.

In the above activity, while some honey sticks on the paper, all the honey slides down the lotus leaf. This is because the tiny hair structures on the leaf's surface repel the honey (Fig 3.1a).

Nano originates from the Greek word 'nanos' which means dwarf.

The hair structures are in the scale of **nanometre** (nm). Materials in nano scale exhibit some intriguing and strikingly different properties compared with the same materials in bulk structures. Let us learn more about these materials.

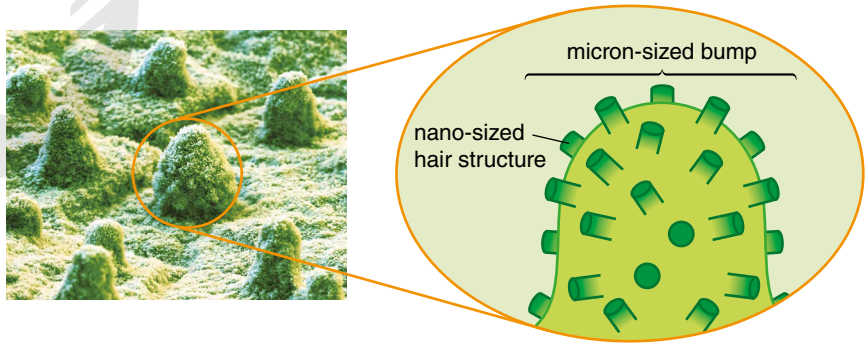


Fig 3.1a *Nano-sized hair structures on the leaf surface.*

nanometre 納米

DSE goal

Understand that nano means 10^{-9}.

Realize that materials in nano scale can exist in various forms, such as nanowires, nanotubes and nanoparticles.

DSE exam

22(2)Q2.8

This definition has been adopted by the European Commission since 2011.

1 Nano materials

Nano (n) is a unit prefix meaning 10^{-9}. Nano materials are usually defined as follows:

Nano materials are those materials with at least one dimension in the range 1 nm to 100 nm, where 1 nm = 10^{-9} m.

Figure 3.1b should give you some idea of how large the nano scale is.

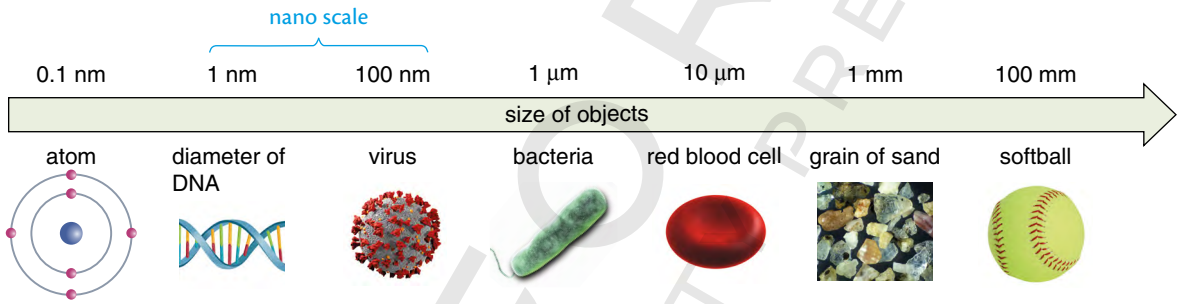


Fig 3.1b *Objects of different sizes.*

Nano materials can be formed naturally or artificially. They can be produced by breaking up a bulk structure or arranging atoms into the desired structure. They exist in various forms.

1 A nanosheet has only one dimension in nano scale (Fig 3.1c).

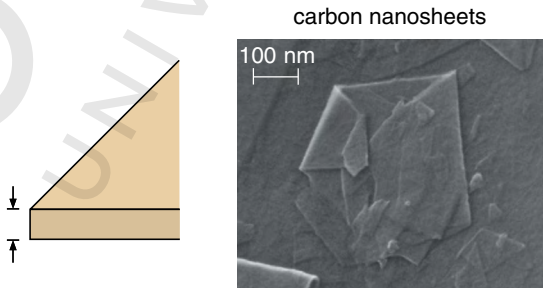


Fig 3.1c *Nanosheets.*

2 A nanotube or a nanowire has two dimensions in nano scale (Fig 3.1d).

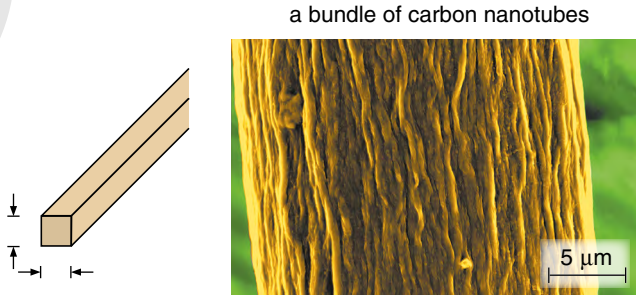


Fig 3.1d *Nanotubes.*

3 A nanoparticle has all three dimensions in nano scale (Fig 3.1e).

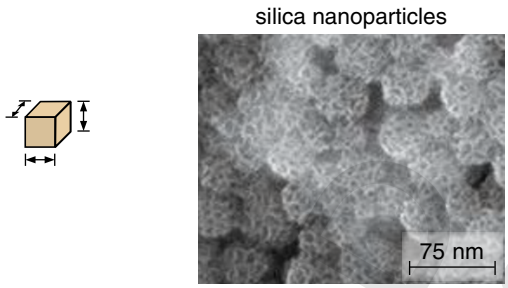


Fig 3.1e *Nanoparticles.*

Carbon is a very good example of nano materials as it exists in all the three forms mentioned above (Table 3.1a).

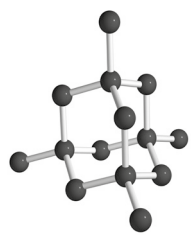
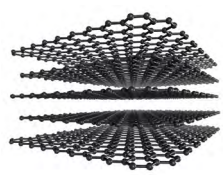
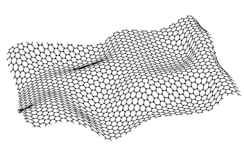
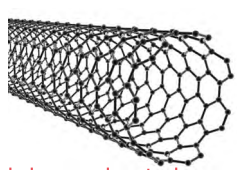
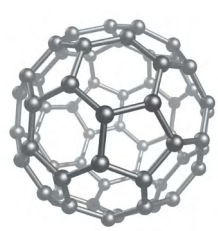
Bulk form	diamond	• atoms arranged in crystal structure • hardest natural material • electrical insulator • best natural thermal conductor ✎ Each carbon atom in diamond is held firmly by four sp^3 bonds with four neighbouring carbon atoms resulting in a giant three dimensional structure which is hard and does not conduct electricity.	
	graphite	• atoms arranged in layers • soft • good electrical and thermal conductor • carbon layers can be easily separated ✎ Each carbon atom in graphite is held firmly by three sp^2 bonds with three neighbouring carbon atoms resulting in giant two dimensional sheets (graphene). These sheets are held loosely together by Van der Waals forces. The delocalized electrons between these sheets make graphite electrically conducting.	
Nanosheet	*graphene*	• atoms arranged in a single layer, can be extracted from graphite • thinnest material • very good electrical and thermal conductor	
Nanotube	*carbon nanotube* (CNT)	• atoms arranged to form a tube, can be in different forms • 100 times stronger than steel of the same size, but only one-sixth as heavy • very good electrical and thermal conductor ✎ Like graphite, carbon nano tubes are formed with sp^2 bonds. The delocalized electrons make them electrically conducting. Due to the work of Sumio Iijima (1939–), carbon nanotubes become widely known and related research is accelerated.	
Nanoparticle	*buckyball*	atoms arranged in a hollow cage, can be in different sizes and different forms ✎ In 1985, a new form of carbon, called the fullerene, or the buckyball, was found. The first discovered fullerene is C_{60}, which consists of 60 carbon atoms. Its structure resembles an old-style football, with a diameter of 1.01 nm. Robert Curl Jr., Sir Harold Kroto and Richard Smalley were awarded the Nobel Prize in Chemistry in 1996 for their discovery of fullerenes.	

Table 3.1a *Different forms of carbon.*

We can see that the same material may have different arrangements of atoms in bulk form and in nano form. This leads to different physical properties.

graphene 石墨烯　　carbon nanotube 納米碳管　　buckyball 布克球

The method that the two scientists used is named as the Scotch tape method. Its procedure is as follows:

Place a thin sample of graphite (with a large number of graphene layers) onto a piece of adhesive tape. Fold the tape and peel it to obtain smaller number of graphene layers. Repeat this many times and a single graphene layer can be obtained.

Know more

Geim and Novoselov

In 2004, *Andre Geim* (Fig a) and *Konstantin Novoselov* (Fig b) isolated individual graphene layers from a piece of graphite by using adhesive tape (Fig c). They were awarded the Nobel Prize in Physics in 2010.

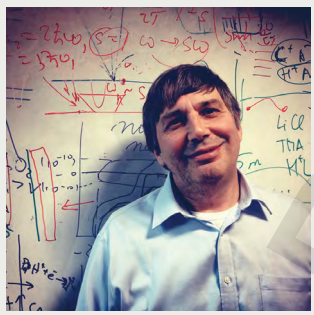


Fig a



Fig b



Fig c

DSE goal

Realize that materials often exhibit different physical properties when their sizes are reduced to nano scale.

DSE exam

12(2)Q2.8, 15(2)Q2.8, 18(2)Q2.8

2 Physical properties of nano materials

A material in nano size often has properties different from that in bulk size. For example, most materials have lower melting points in nano size than in bulk size. Let us take a look at some more properties of nano materials.

For example, the melting points of gold in bulk size and in the form of 1-nm nanoparticles are 1064 °C and 20 °C respectively.

a Electrical properties

The electrical conductivity does not change when you change the diameter of a material in bulk size or when you twist a conducting wire. However, in the case of carbon nanotubes, conductivity changes with its diameter. Its conductivity also changes when it is twisted.

Copper is another example. It is a very good conductor in bulk size. However, its conductivity drops drastically when reduced to nano size.

Andre Geim 海姆　Konstantin Novoselov 諾沃肖洛夫

b Mechanical properties

Traditional non-metals are generally weak and soft in bulk scale. However, some materials become strong and hard when reduced to nano scale. For example, carbon nanotubes are a hundred times stronger than steel.

c Optical properties

The colour of a material may change when it is reduced to nano size. For example, gold nanoparticles of different sizes show different colours. Since the 10th century, glass makers have been able to make stained glass of different colours by adding gold in the glass-making process (Fig 3.1f).

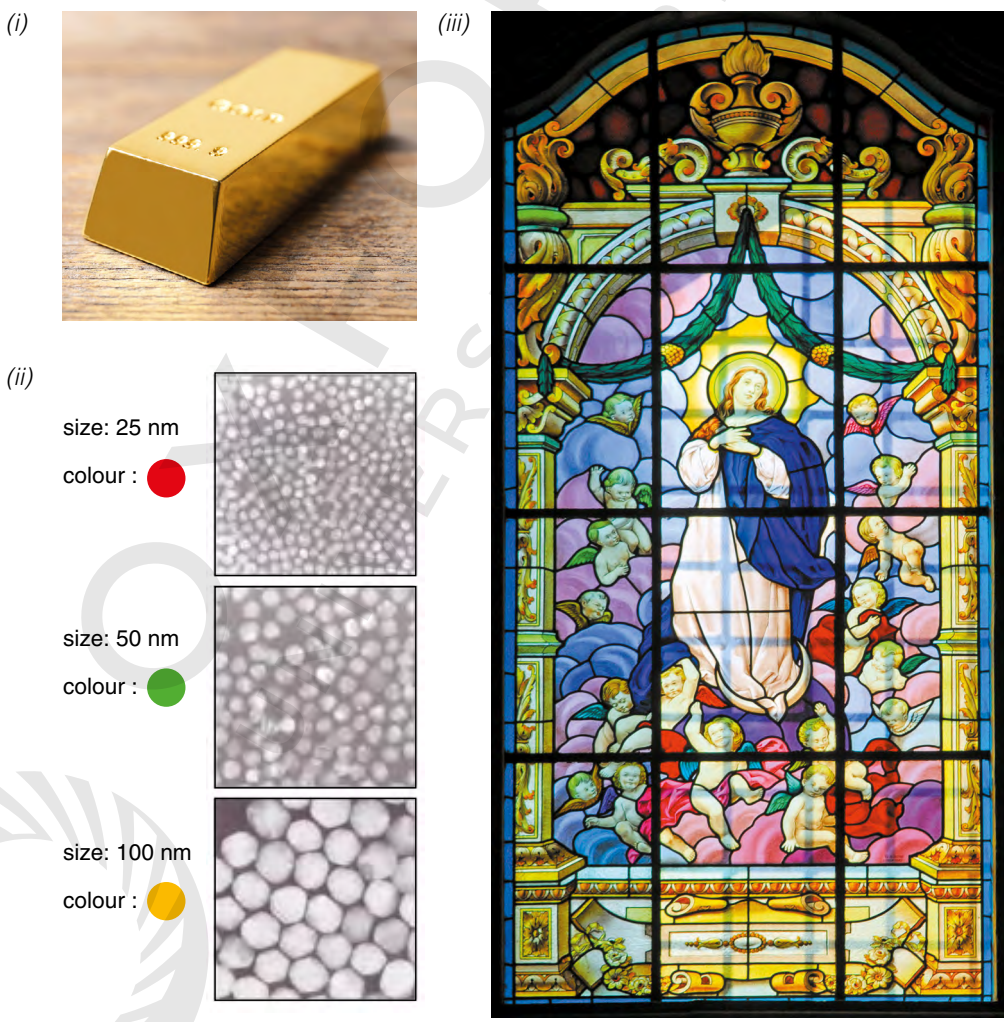


Fig 3.1f *(i) Gold in bulk form, (ii) the colours of gold nano particles of different sizes, (iii) stained-glass window.*

Titanium dioxide (TiO_2) and zinc oxide (ZnO) are two other examples. These two materials are effective in blocking ultra-violet radiation and are commonly used in sunscreens. They are white in bulk size but are transparent when reduced to nano size. A sunscreen with these nanoparticles is transparent and does not whiten the skin when applied.

Google Forms Microsoft Forms

When materials are reduced to nano scale in size, they may exhibit different electrical (e.g. conductivity), mechanical (e.g. strength and hardness) and optical (e.g. colour and transparency) properties.

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 Which of the following properties of a material may change when its size is reduced to nano scale? Put a tick in the corresponding box(es).

- ☐ Colour
- ☐ Hardness
- ☐ Electrical conductivity
- ☐ Melting point

2 *True or False:* ☆ HKDSE 2018 Paper 2 Q2.8

(a) Diamond is a poor thermal conductor and a poor electrical conductor as it does not have free electrons. (T/F)

(b) Buckyballs are formed by carbon atoms arranged in hollow cages. (T/F)

(c) A nano material must have at least one dimension shorter than 1 nm. (T/F)

DSE exam

13(2)Q2.8, 19(2)Q2.8, 20(2)Q2.8, 21(2)Q2.8, 23(2)Q2.8

3 Why do nano materials behave differently?

For simplicity, we do not distinguish atoms, molecules and ions in a material.

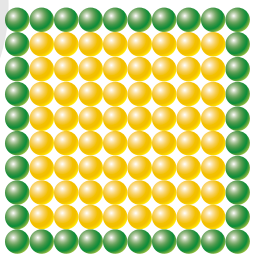
▶ Besides the different arrangements in atoms, the surface effect and the quantum effect are also the reasons for nano materials behaving differently from their bulk structures.

a Surface effect

As the size of an object decreases, its surface-area-to-volume ratio increases. Therefore, the percentage of atoms at the surface increases. Figure 3.1g shows an analogue in 2-dimension.

A material in nano scale is also more efficient in acting as a catalyst due to the large surface-area-to-volume ratio.

▶ Usually, only the atoms located at the surface of an object react with the environment. Therefore, a material can become much more reactive in nano scale than in bulk scale.



total no. of atoms: 100
atoms at surface: 36 (36%)



total no. of atoms: 16
atoms at surface: 12 (75%)

Fig 3.1g *An analogue in 2-dimension: when the size decreases, the percentage of atoms at the surface increases.*

b Quantum effect

As studied in Chapter 2, classical physics fails to explain many phenomena in the regime of atomic scale. They require a new set of physics, called *quantum mechanics*, to explain. Similarly, the very different electrical, mechanical and optical properties of nano materials require quantum mechanics to explain.

Quantum mechanics is out of the scope of this book.

Example 1 Surface-area-to-volume ratio

(a) Find the surface-area-to-volume ratio R_1 of a cube of side length 1 cm.

(b) Find the surface-area-to-volume ratio R_2 of a cube of side length 1 nm.

(c) Find $R_2 : R_1$.

Solution

Tip
A cube has 6 surfaces.

(a) $R_1 = \dfrac{\text{total surface area}}{\text{volume}} = \dfrac{6 \times (0.01)^2}{(0.01)^3} = 600 \text{ m}^{-1}$

(b) $R_2 = \dfrac{\text{total surface area}}{\text{volume}} = \dfrac{6 \times (1 \times 10^{-9})^2}{(1 \times 10^{-9})^3} = 6 \times 10^9 \text{ m}^{-1}$

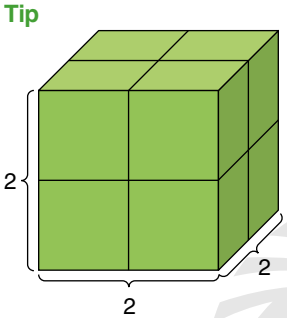
(c) $R_2 : R_1 = 6 \times 10^9 : 600 = 10^7 : 1$

Example 2 Dividing a cube

A cube of side length 1 cm is divided into cubes of side length (a) 0.5 cm, (b) 1 nm. What is the ratio of the total surface area after dividing to that before dividing in each case?

Solution

Tip

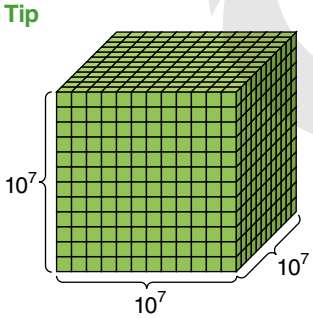


(a) Number of cubes on each side $= \dfrac{1 \text{ cm}}{0.5 \text{ cm}} = 2$

$\therefore$ Total number of cubes after dividing $= (2)^3 = 8$

Ratio of total surface area $= \dfrac{6(0.005)^2 \times 8}{6(0.01)^2} = 2$

Tip



(b) Number of cubes on each side $= \dfrac{1 \text{ cm}}{1 \text{ nm}} = \dfrac{0.01 \text{ m}}{10^{-9} \text{ m}} = 10^7$

$\therefore$ Total number of cubes after dividing $= (10^7)^3 = 10^{21}$

Ratio of total surface area $= \dfrac{6(10^{-9})^2 \times 10^{21}}{6(0.01)^2} = 10^7$

quantum mechanics 量子力學

Quick check 2

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 *True or false:*

(a) When the size of an object increases, the percentage of particles at the surface decreases. (T/F)

(b) A material can become much more reactive when its size is reduced to nano scale. (T/F)

(c) In the nano scale, the quantum effect should be considered. (T/F)

2 A material is broken up into nano size particles without changing the total volume. Determine whether the following statements are true.

(a) The total surface area increases. (T/F)

(b) The chemical properties may change. (T/F)

(c) The physical properties may change. (T/F)

(d) The material may become more reactive. (T/F)

Learn by practice 1

1 A cube of side length 10 cm is divided into small cubes of side length 10 nm.

(a) Find the number of small cubes formed.

(b) Find the surface-area-to-volume ratio of the original cube.

(c) Find the surface-area-to-volume ratio of the small cube.

(d) By how many times does the total surface area of the cube increase after being divided?

DSE goal

Describe recent developments and applications of nanotechnology in various areas related to daily life.

DSE exam

12(2)Q2.7, 14(2)Q2.8, 16(2)Q2.8

4 Applications and developments

a Health care

Under ultra-violet radiation, titanium dioxide (TiO_2) speeds up reactions that give detoxicating and antibacterial effects. Sprays containing TiO_2 nanoparticles can be used on different surfaces to inactivate viruses (including COVID-19) (Fig 3.1h).

TiO_2 acts as a *photocatalyst*, which is a catalyst that works under light (UV radiation). It is in nano form to increase the surface area.



Fig 3.1h *TiO_2 nanoparticles can be used on different surfaces to inactivate virus.*

photocatalyst 光觸媒

Know more

Molecular machines

In 2016, *Jean-Pierre Sauvage, Fraser Stoddart* and *Ben Feringa* were awarded the Nobel Prize in Chemistry for their pioneering works in developing molecular machines.

▸ p.134

Nanoparticles have many other uses in the medical field.

1 Silver nanoparticles have antimicrobial effects (Fig 3.1i). They can be used as an antimicrobial agent in the treatment of wounds.

2 Some nano materials are used as chemical and biological sensors. For example, cadmium selenide nanoparticles can be injected into patients' bodies for detection of early-stage cancers.

3 New drugs and therapies based on nanotechnology are being developed. These drugs can reach the target locations in the body with great precision. In the future, there may even be nano robots injected into patients' bodies to deliver drug, repair cells and carry out other medical tasks (Fig 3.1j).

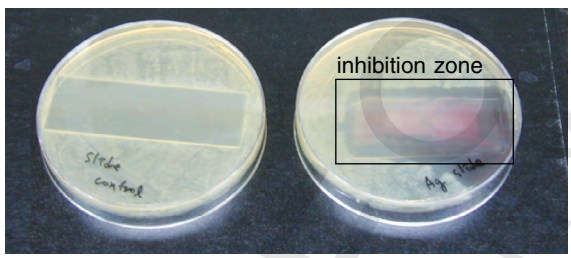


Fig 3.1i *Glass slide with silver nanoparticle coating inhibits growth of E. coli (right) compared with the control (left).*

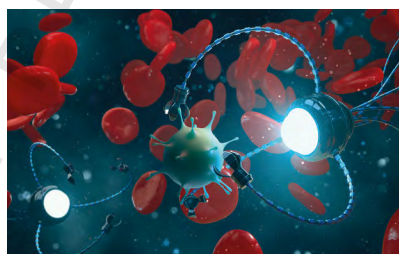


Fig 3.1j *Computer artwork of a nano robot.*

→ This video demonstrates the Lotus effect.

Video

Due to the same effect, honey on a lotus leaf slides down easily, as you saw in the activity on p.96.

b Materials

✎ Water-repelling property is sometimes referred to as hydrophobic. 'Hydro' means water. 'Phobic' means fear of something.

The nano-sized hair structures on a lotus leaf repel water. This effect helps keep the leaf clean. When water falls on the leaf, it beads up, rolls off and picks up the dirt particles along with it (Fig 3.1k). This self-cleaning property is known as the **Lotus effect**.

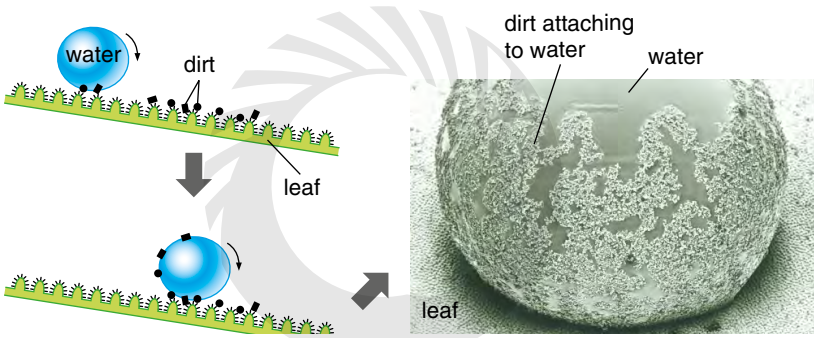


Fig 3.1k *Self-cleaning by repelling water.*

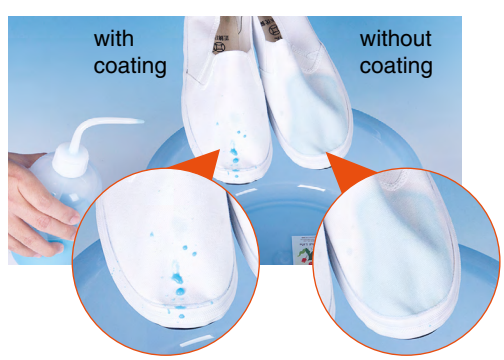


Fig 3.1l *Shoes with and without nano-coating.*

Video

→ This video shows a stain-resistant shirt.

Nano-coatings resembling the Lotus effect have been made. For example, a shoe can become water-repelling and self-cleaning by spraying nano-coating onto it (Fig 3.1l). Other applications include self-cleaning windows, stain-repelling fibres, anti-graffiti coatings for walls, etc.

E. coli 大腸桿菌　Jean-Pierre Sauvage 索瓦日　Fraser Stoddart 斯托達特
Ben Feringa 費林加　Lotus effect 蓮花效應

Video

→ This video shows the water-attractive property of a self-cleaning glass.

Nano structures can also be water-attractive. For example, a glass coated with a thin film (about 15 nm-thick) of titanium dioxide (TiO_2) attracts water. This gives the glass a self-cleaning effect (Fig 3.1m).

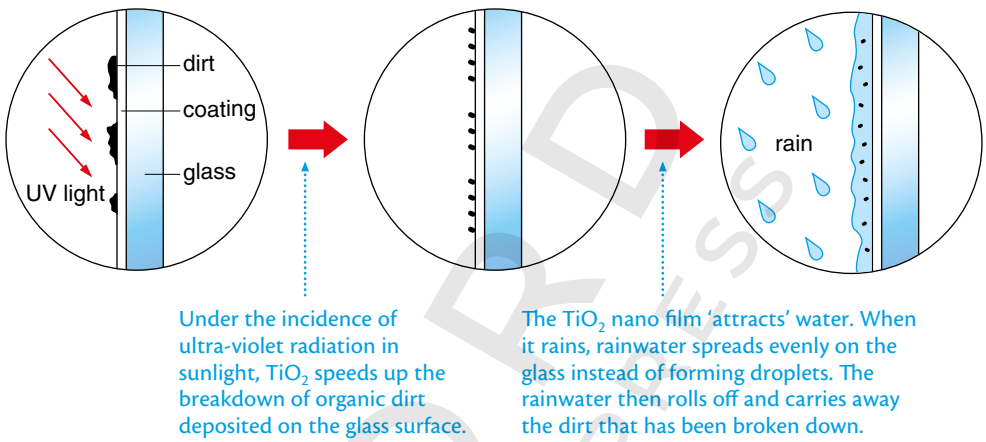


Fig 3.1m *Self-cleaning by attracting water.*

Know more

Future uses of CNTs

CNTs may be good building materials for cars and aeroplanes in the future. Lighter vehicles would mean better fuel efficiency and increased strength would help improve passenger safety. Scientists are also exploring the possibility of building a space elevator using CNTs.

Nano materials can also be added to materials to enhance their mechanical properties. For example, carbon nanotubes (CNTs) are lighter than aluminium but a hundred times stronger than steel. They are combined with carbon fibres to produce lighter and stronger sports equipment, including bicycles and tennis racquets (Fig 3.1n).



Fig 3.1n *Bicycles and tennis racquets can be made lighter and stronger with the use of CNTs.*

C Information technology

Processors (Fig 3.1o) are responsible for data storage and computation in electronic equipment. With the development of nanotechnology, a processor with a much higher storage capacity and computing performance can be achieved.

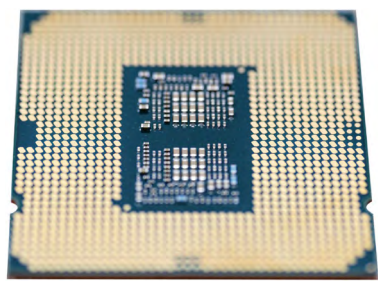


Fig 3.1o *A processor.*

d Environment

Silver nanoparticles have detoxicating, deodorizing and antibacterial properties. They are widely used in air and water treatment facilities (Fig 3.1p).



Fig 3.1p *Silver nanoparticles are widely used in air conditioner filters.*

The usage of nano materials also helps the development in clean energy. Figure 3.1q shows how the efficiency of a wind turbine could be improved. Scientists also showed that the efficiency of silicon solar cells can be enhanced by coating silver nanoparticles on them.

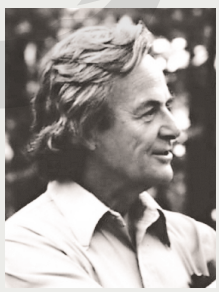


Fig 3.1q *Nanotechnology could improve the efficiency of a wind turbine.*

The script of Feynman's talk can be found in the following website:

More information about Feynman's life can be found in the following website:

Know more



Idea of nanotechnology

The idea of nanotechnology was first introduced by *Richard Feynman* in his talk 'There's Plenty of Room at the Bottom'. Feynman was awarded the Nobel Prize in Physics in 1965 for his contribution in another field of science (*quantum electrodynamics*).

Richard Feynman (1918–1988).

OUP eResources

Richard Feynman 費曼　　quantum electrodynamics 量子電動力學

DSE goal

Discuss potential hazards, issues of risks and safety concerns for our lives and society in using nanotechnology.

5 Potential hazards

Despite a lot of promising applications offered by nanotechnology, there are concerns about the potential risks. A substance which is harmless at an ordinary size may become harmful to public health and the environment in nano size.

a Health risks

Nano materials can enter the human body in three ways (Fig 3.1r):

1 Free nano materials may be released during manufacturing processes or through *abrasion* of products. They may be inhaled into the lungs or may pass through the skin.

2 Nano materials in products that are in direct contact with us (such as sunscreens and cosmetics) may penetrate our skin.

3 Nano materials in food or beverages may enter our body when eaten.

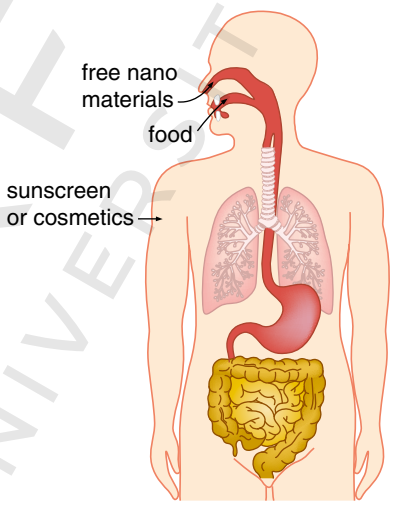


Fig 3.1r *Nano materials may enter the human body.*

Some experiments have shown that some nano materials may be toxic to humans. However, scientists are still unable to come to a conclusion. The following lists some possible hazards, based on experiments:

1 When inhaled into lungs, some nano materials (such as CNTs and ZnO nanoparticles) may cause damage to the lungs. TiO_2 nanoparticles has even been classified as 'possible carcinogenic to humans'.

2 Nano materials may enter the bloodstream after being inhaled. Once in the blood, they may enter other vital organs (such as the brain) and cause damage there.

3 Some nano materials may facilitate the production of reactive molecules that may lead to cell damage.

abrasion 磨損

b Environmental impact

Free nano materials can easily spread through the atmosphere and contaminate water, air, soil and ground water. Some of them, such as buckyballs and silver nanoparticles, are toxic. Some may accumulate in living organisms, enter the food chain and damage the ecosystem. They may also become a new class of non-biodegradable pollutants.

c Hopes and concerns

Nanotechnology has become a controversial issue. Table 3.1b lists some arguments supporting and opposing nanotechnology.

For	Against
• Nanotechnology has huge potential to benefit drug development, water decontamination, information and communications technologies, and the development of cheap sustainable energy. • Nanotechnology may also benefit the environment. For example, new materials which are rapidly biodegradable are now being developed. • The manufacture of nano products generally uses less resources and produces less waste.	• Since little detail is yet known about its effects, the use of free nano materials in environmental applications should be prohibited until proven safe by appropriate research. • Nanoparticles in consumer goods may be released into the environment and cause unintended consequences. For example, silver nanoparticles released from odourless socks may destroy bacteria which are crucial to the environment.

Table 3.1b *Arguments supporting and opposing nanotechnology.*

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) A nano material is safe to use in any situation if the material in bulk form is non-toxic. (T/F)

(b) After being inhaled, nano materials may enter the bloodstream and cause damage to human organs. (T/F)

(c) Nanoparticles may accumulate in living organisms and enter the food chain. (T/F)

2 Which of the following are the applications of nanotechnology? Put a tick in the corresponding boxes.

- ☐ Killing virus
- ☐ Enhancing mechanical strength of materials
- ☐ Keeping materials clean

Exercise 3.1

★**1** By which of the following ways may nanoparticles enter our body?

(1) Free nanoparticles may be inhaled into our lungs.

(2) Nanoparticles in sunscreen may penetrate our skin.

(3) Nanoparticles in food may be absorbed by our digestive system.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★**2** Which of the following are the effects of TiO_2 nanoparticles?

(1) Water-repelling

(2) Inhibiting the growth of bacteria

(3) Blocking UV radiation

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★**3** A shirt with water-repelling and stain-resistant properties is shown (Fig a).

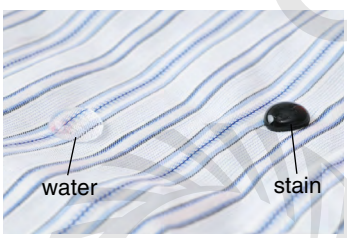


Fig a

This can be achieved by

(1) adding nanoparticles of ZnO to the fabric.

(2) adding nano-coatings with Lotus effect to the fabric.

(3) adding carbon nanotubes to the fabric.

A (2) only

B (3) only

C (1) and (2) only

D (1) and (3) only

4 Give an example of a nano material which has

(a) one dimension,

(b) two dimensions,

(c) three dimensions

in nano scale.

★**5** A cube of side length L is divided into 10^{18} identical nano-sized cubes. Find the total surface area of the nano-sized cubes in terms of L.

★**6** Some contact lens cases (Fig b) are incorporated with silver nanoparticles. Suggest how this usage of nanotechnology improves the lens cases.



Fig b

★**7** Suggest possible ways, with the aid of nanotechnology, to improve

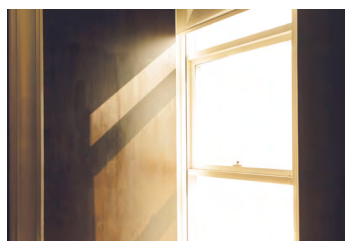
(a) an umbrella so that it is waterproof,



(b) a tennis racquet so that it is lighter and stronger,



(c) a window glass so that it can block UV radiation.



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3.2 Seeing at nano scale

If there is a water droplet on the display of a smartphone, you may see some discrete dots through the droplet (Fig 3.2a). Why can't you see the dots when there is no water droplet?

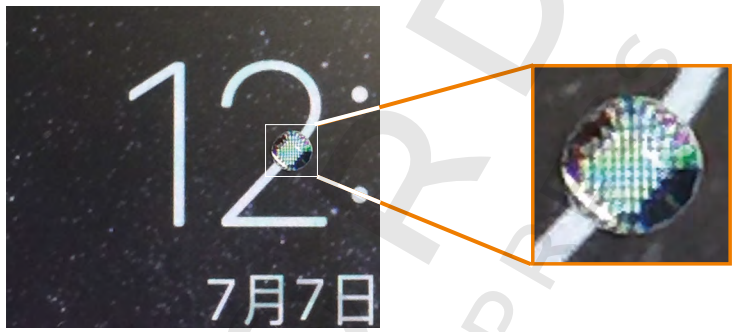


Fig 3.2a *A water droplet on the display of a smartphone.*

The answer is related to the ability of the eye to distinguish two points, i.e. the **resolving power**, or the **resolution**, of the eye. This resolving power is limited by the diffraction of light.

DSE goal

Describe the limitations of optical microscope in seeing substances of small scale.

DSE exam

15(2)Q2.4, 16(2)Q2.6, 17(2)Q2.6, 17(2)Q2.7, 19(2)Q2.7, 22(2)Q2.5

Cross-link

You learned diffraction of light in Book 3B Ch 6.1. The reason for the formation of the bright and dark fringes is beyond the scope of this book.

1 Diffraction-limited resolving power

When light passes through a narrow slit or a small hole, diffraction occurs (Fig 3.2b).

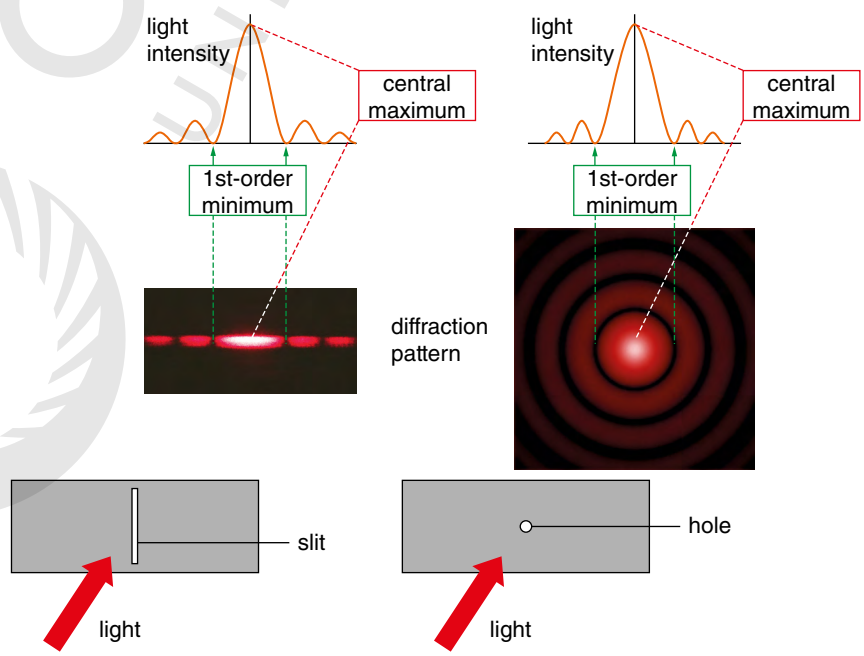


Fig 3.2b *Diffraction patterns formed when light passes through a narrow slit or a small hole.*

resolving power 分辨率　　resolution 解像度

Experiment 3a

Resolving power of the human eye (p.148)

Video

→ This video demonstrates Expt 3a.

The same phenomenon occurs when light enters an optical instrument (e.g. a microscope) or an eye through its opening, which is usually called the *aperture*. When light from two points *X* and *Y* passes through the aperture, two diffraction patterns will be formed (Fig 3.2c).

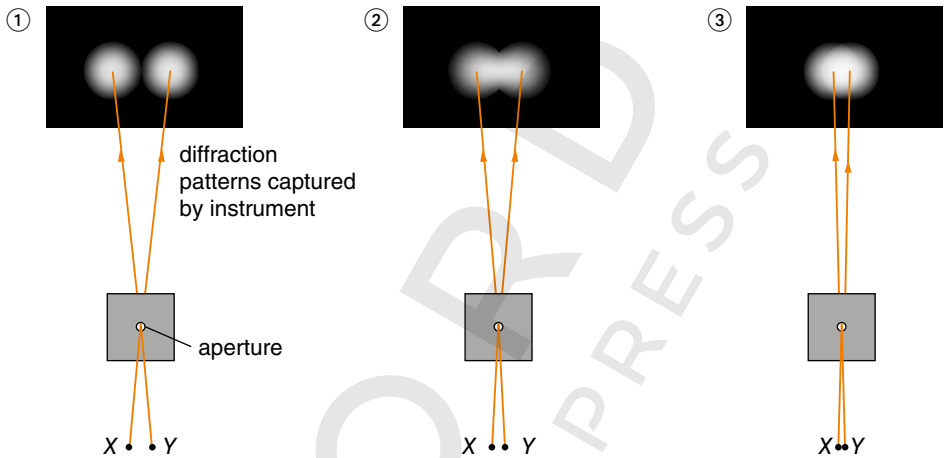


Fig 3.2c *Light from two points passing through the aperture.*

Simulation

→ This simulation demonstrates the conditions for two points that can be resolved. It is available on OUP web in PC version (.exe) only.

① If *X* and *Y* are far apart, the two diffraction patterns will not overlap, and the two points can be distinguished by the instrument. The two points are said to be resolved.

② If *X* and *Y* are closer, the diffraction patterns become partially overlapped and less readily to be resolved.

③ If *X* and *Y* are very close, most of the diffraction patterns will overlap and the two points are no longer resolved.

To determine the criterion for two points to be 'just resolved', we can refer to the **Rayleigh criterion** (Fig 3.2d):

Two points can just be resolved when the central maximum of the diffraction pattern of one point is centred on the first-order minimum of the diffraction pattern of the other.

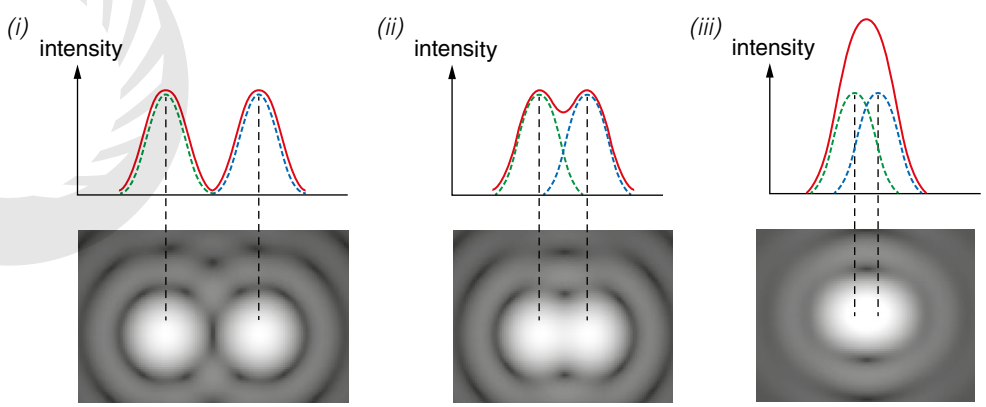
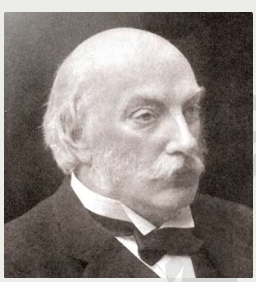


Fig 3.2d *According to Rayleigh criterion, two objects are (i) resolved clearly, (ii) just resolved, and (iii) not resolved.*

Know more

John Strutt (1842–1919)



John Strutt is the 3rd Baron Rayleigh. He defined the Rayleigh criterion and made major contributions to other fields of physics. He was awarded the Nobel Prize in Physics in 1904 for the discovery of the element argon.

aperture 孔徑 Rayleigh criterion 瑞利判據 John Strutt 斯特拉特

The Rayleigh criterion can be expressed as a formula to find θ_R, which is the **minimum** **angular separation** between two points that can be resolved (Fig 3.2e). θ_R is also known as the **angular resolution** of the instrument. It is measured in **radians**.

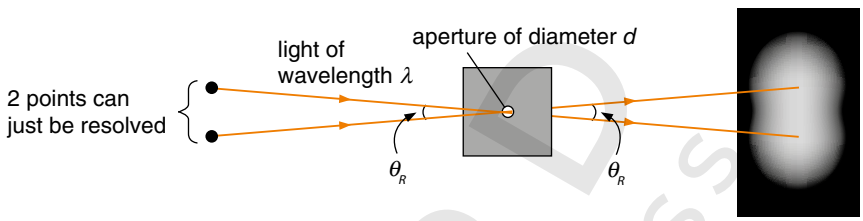


Fig 3.2e *θ_R is the angular separation between two points which can just be resolved.*

$$\text{angular resolution (rad)} \quad \theta_R \approx \frac{1.22\lambda}{d} \quad \begin{matrix}\text{wavelength of light (m)} \\ \text{diameter of aperture (m)}\end{matrix}$$

A smaller value of θ_R means that two points closer together can be resolved.

→ The effect of aperture size on resolving power can be illustrated by a set of Saturn images taken by telescopes with various aperture sizes. See Figure 3 in the following webpage:

▶ The **smaller** the value of θ_R, the **higher** the resolving power. Therefore, decreasing the wavelength of light or increasing the size of the aperture can increase the resolving power of an optical instrument.

This can be understood by considering the factors affecting diffraction. When the wavelength decreases or the gap width (the size of aperture) increases, diffraction is less significant, so two points can be resolved more easily (Fig 3.2f). In other words, the resolving power increases.

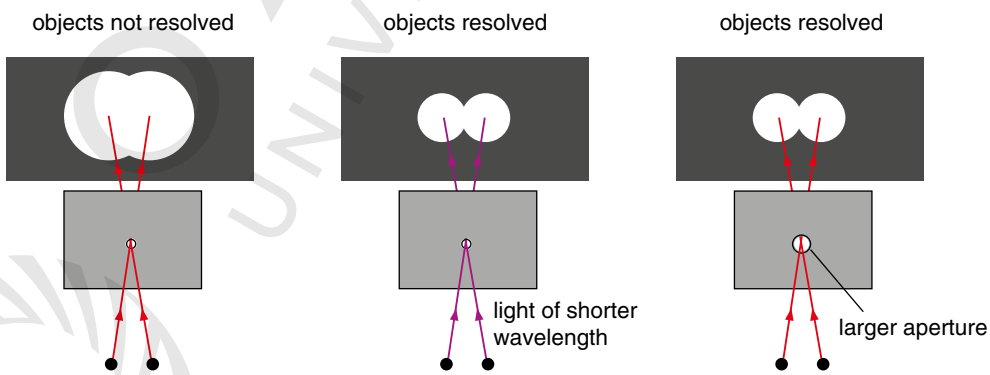


Fig 3.2f *Two points can be resolved more easily when diffraction is less significant.*

If two points can just be resolved and their distance from the aperture is r (Fig 3.2g), the distance s between the two points is given by

$$s \approx r\theta_R$$

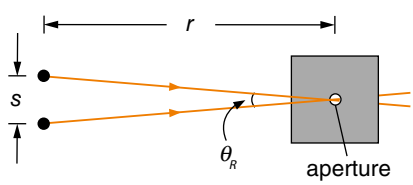


Fig 3.2g *Separation s and angular separation θ_R of two points.*

Skill *Mathematics*

Approximation of the arc length ▶ p.135

angular separation 角間距　　angular resolution 角解像度

Example 3 | Resolving power of the eye

Stuart's eye is 25 cm from an object. The wavelength of the light reflected from the object is 500 nm and the diameter of Stuart's pupil is 5 mm.

(a) Estimate the angular resolution θ_R of Stuart's eye by considering the Rayleigh criterion.

(b) Estimate minimum distance s between two points on the object which can be resolved by Stuart's eye.

(c) If Stuart is further away from the object, how do θ_R and s change?

Solution

(a) $\theta_R \approx \dfrac{1.22\lambda}{d} = \dfrac{1.22(500 \times 10^{-9})}{5 \times 10^{-3}} = 1.22 \times 10^{-4}$ rad

This is the theoretical limits only. In reality, the resolving power of the eye could be reduced (θ_R larger) due to eye defects and air pollution. ▶

(b) $s \approx r\theta_R = (0.25)(1.22 \times 10^{-4}) = 3.05 \times 10^{-5}$ m

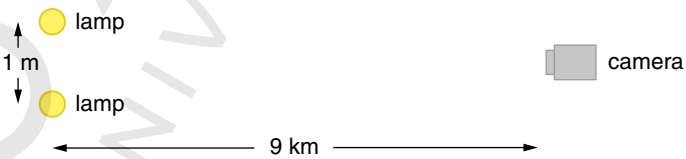
This is about the diameter of a human hair. ▶

(c) θ_R remains unchanged as it is independent of r.
s increases as r increases according to $s \approx r\theta_R$.

Two points that can just be resolved are further apart. ▶

Example 4 | Resolving power of a camera

Two lamps 1 m apart emits light of wavelength 590 nm. A camera is 9 km away. Its aperture has a diameter of 6.2 mm.



(a) What is the angular resolution θ_R of the camera according to the Rayleigh criterion?

(b) The lamps cannot be resolved by the camera. Explain.

(c) Suggest one method that allows the lamps to be resolved by the camera.

Solution

(a) $\theta_R \approx \dfrac{1.22\lambda}{d} = \dfrac{1.22(590 \times 10^{-9})}{6.2 \times 10^{-3}} = 1.16 \times 10^{-4}$ rad

(b) Minimum distance s that can be resolved by the camera
$= r\theta_R = (9000)(1.16 \times 10^{-4}) = 1.04$ m

Since the distance between the lamps is shorter than 1.04 m, they cannot be resolved by the camera.

We can also consider the angular separation θ of the two lamps: ▶

$\theta \approx \dfrac{s}{r}$

$= \dfrac{1}{9000}$

$= 1.11 \times 10^{-4}$ rad

$< 1.16 \times 10^{-4}$ rad

(c) Move the camera closer to the lamps. / Increase the distance between the lamps.

Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) An optical instrument with a larger angular resolution θ_R has a higher resolving power. (T/F)

(b) The resolving power of an eye is higher if the pupil becomes larger. (T/F)

(c) The resolving power of an optical instrument is higher if light of shorter wavelength is used. (T/F)

2 Two lamps can just be resolved by John. Complete the table below.

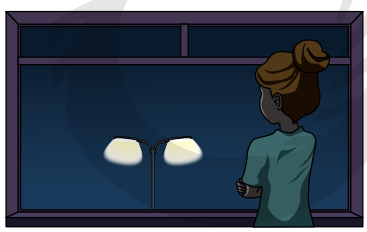
Action	Change in angular separation between lamps	Can the lamps be resolved by John?
(a) Place the lamps closer to each other		
(b) John walks towards the lamps		
(c) Increase the brightness of the lamps		

3 By considering the Rayleigh criterion, complete the following table about the resolution of an instrument.

	θ_R / rad	λ / m	d / m	s / m	r / m
(a)	2×10^{-4}	6×10^{-7}		3×10^{-5}	
(b)		3×10^{-8}	6×10^{-3}		0.01

Learn by practice 2

1 A girl looks at the two lamps on a lamp post at night. The diameter of the girl's pupil is 6 mm. The lamps are 1 m apart from each other and give out light of wavelength 590 nm.



(a) Estimate the angular resolution θ_R of the girl's eye by considering the Rayleigh criterion.

(b) The girl can just resolve the lamps. Find her distance from the lamp post.

2 An eagle is flying at a height of 200 m. The diameter of its pupil is 7 mm. Two beans A and B on the ground are 2 cm apart. They reflect light of wavelength 550 nm.

(a) Estimate the angular resolution θ_R of the eagle's eye by considering the Rayleigh criterion.

(b) Can A and B be resolved by the eagle?

(c) Can A and B be resolved by the eagle when

(i) the distance between A and B becomes 1.5 cm;

(ii) the eagle flies to a lower position?

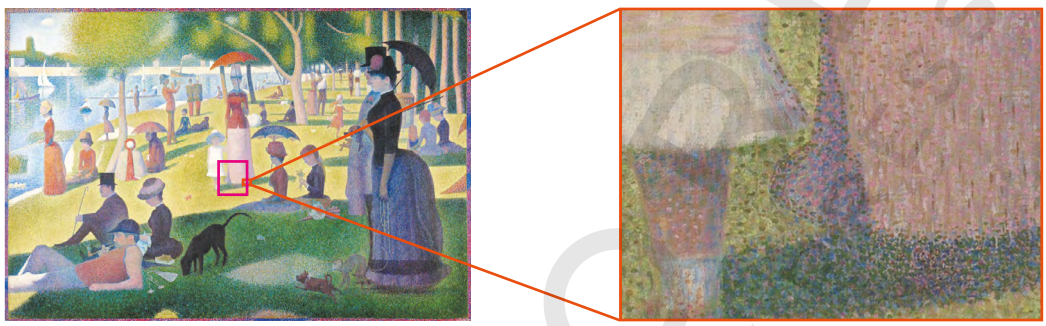
☆ HKDSE 2019 Paper 2 Q2.7

→ This videos demonstrates how this question is solved.

DSE practice 1 | Art and physics

Video

The French artist *Georges Seurat* was famous for developing a technique of painting called *pointillism*, in which dots of colour are placed closely together on the *canvas*. The figure below shows his famous painting using this technique. A person looking at the painting from a distance cannot distinguish the dots. The colour perceived is a result of mixing the colours of nearby dots.



A Sunday on La Grande Jatte—1884, Georges Seurat.

Kate walks towards the painting. She can just distinguish two violet dots separated by a distance s when she is at a distance D from the painting. How should she move such that she can just distinguish two red dots separated by the same distance s? The diameter of her pupils remains unchanged.

A Move about $0.43D$ towards the painting

B Move about $0.57D$ towards the painting

C Move about $0.43D$ away from the painting

D Move about $0.57D$ away from the painting

Solution

$$s \approx r\theta_R$$

$$\approx r \times \frac{1.22\lambda}{d}$$

$$r \approx \frac{sd}{1.22\lambda}$$

$\therefore$ r is approximately inversely proportional to λ (for constant s and d).

$$\therefore \frac{r_R}{r_V} \approx \frac{\lambda_V}{\lambda_R}$$

$$r_R \approx \frac{\lambda_V}{\lambda_R} \times r_V$$

$$\approx \frac{400}{700} \times D$$

$$= 0.57D$$

Distance moved $= D - 0.57D = 0.43D$

$\therefore$ The answer is A.

Common mistake ⚠
Students may not remember that the wavelengths of red light and violet light are about 700 nm and 400 nm respectively.

Common mistake ⚠
Students may not be aware that they should find the distance that Kate moves.

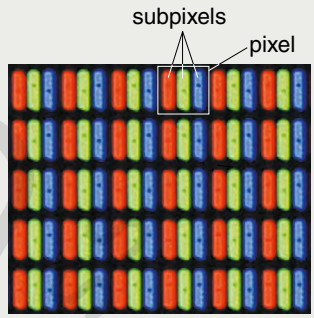
▶ Revision 3 Q14 (p.131)

Georges Seurat 秀拉　pointillism 點描法　canvas 畫布

Know more

Pixels on a smartphone

The display of a smartphone consists of millions of light dots called pixels. Each pixel consists of red, green and blue subpixels. The pixels are so close together that our eye is unable to resolve them. The colours that we see is a result of mixing the colours of nearby subpixels. By controlling the amount of light emitted from each subpixel, the whole range of colours can be produced.



Refer to Figure 3.2a on p.110. The water droplet on the phone acts like a magnifying glass and produces an enlarged image. If the image is large enough, the pixels will be resolved, so discrete light dots, or even the subpixels, can be seen.

2 Optical microscopes

Figure 3.2h shows an optical microscope commonly used in schools. It uses convex lenses to produce an enlarged image.

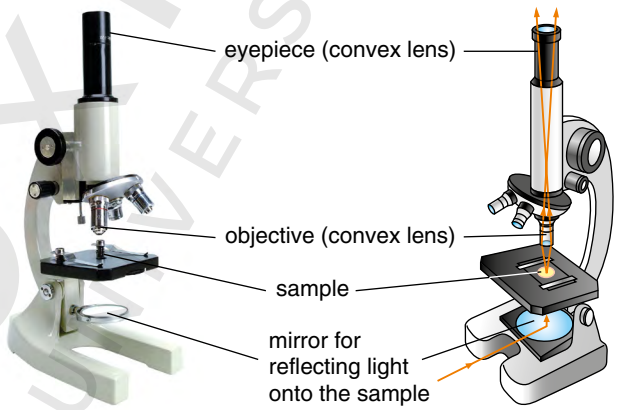


Fig 3.2h *Optical microscope.*

Like other optical instruments, its resolving power is limited by the Rayleigh criterion. Let us estimate its resolving power. Suppose its aperture (the objective lens) has a diameter of 6 mm and is 1 cm away from the sample. Take the wavelength of light as 500 nm. The shortest distance s between two points on the sample that can be resolved is:

$$s \approx r\theta_R$$
$$\approx r \times \frac{1.22\lambda}{d}$$
$$= 0.01 \times \frac{1.22(500 \times 10^{-9})}{6 \times 10^{-3}}$$
$$= 1.02 \times 10^{-6}\ \text{m}$$

We can see that its resolving power is not high enough to observe nano materials. To make a microscope with a higher resolving power, we can either **increase *d*** or **reduce *λ***. A laboratory microscope cannot be too large, so *d* cannot be increased considerably. Therefore, *λ* becomes the most important factor. This realization prompted the invention of the **transmission electron microscope** (TEM).

DSE goal

Describe how a TEM works.

Draw the analogy between the use of electric and magnetic fields in TEMs and lenses in optical microscopes.

Estimate the anode voltage needed in a TEM to accelerate electrons achieving wavelengths of the order of atomic size.

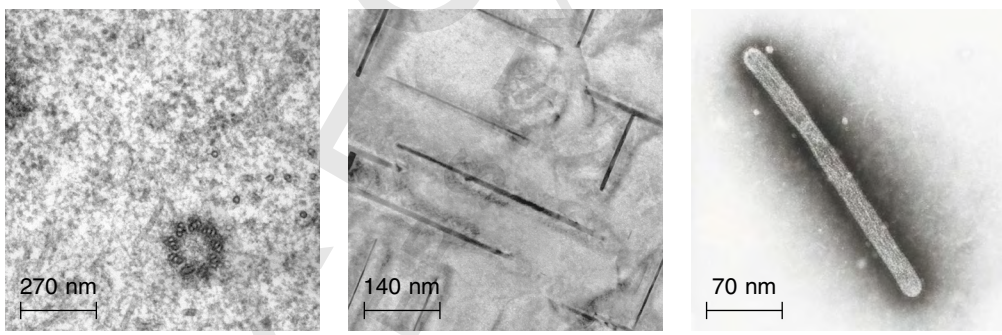
Explain the advantage of high resolution of TEM using Rayleigh criterion for minimum resolvable detail.

DSE exam

13(2)Q2.7, 14(2)Q.2(a)–(b), 15(2)Q2.7, 16(2)Q2.7, 20(2)Q2.6, 21(2)Q2.7, 22(2)Q2.6, 23(2)Q2.7, etc.

3 Transmission electron microscopes

A TEM works like an optical microscope but it uses fast-moving electrons instead of light to form images. Since the wavelength of the electrons is much shorter than visible light, the resolving power of a TEM is much higher than an optical microscope according to the Rayleigh criterion. In practice, a TEM can resolve detail of size down to 0.1 nm (Fig 3.2i).



(i) Brain tissue of mouse (35 000×). *(ii) Al-Cu alloy (68 000×).* *(iii) An H5N1 virion, a type of bird flu virus (150 000×).*

Fig 3.2i *Images taken with TEMs.*

Figure 3.2j shows the basic structure of a TEM.

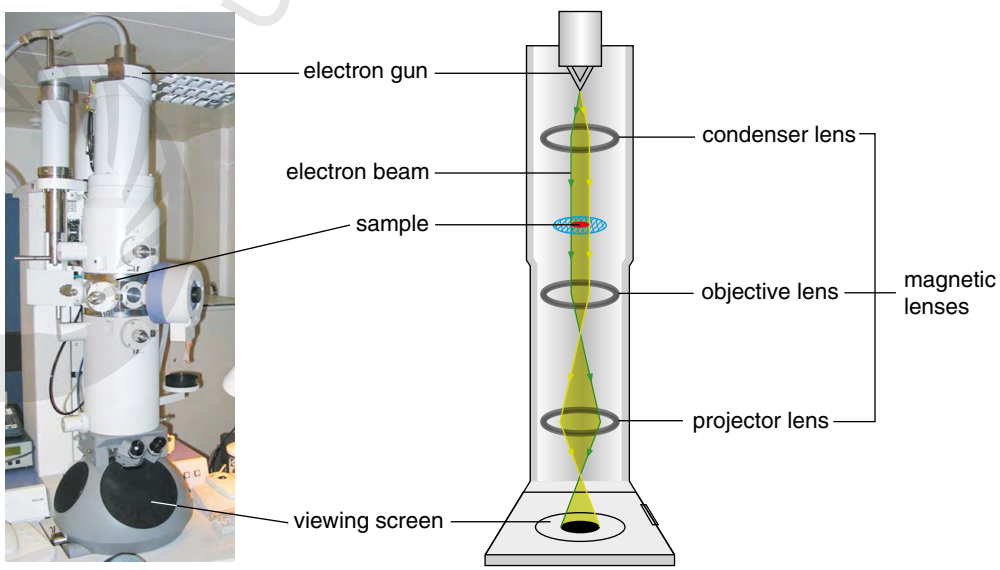


Fig 3.2j *The basic structure of a TEM.*

transmission electron microscope 透射電子顯微鏡

The sample is about a few hundred nanometres thick. It may be damaged by the electrons.

An electron beam is emitted from the electron gun at the upper end. It is focused by the magnetic condenser lens and is transmitted through a very thin sample. The transmitted beam is then projected by the magnetic objective and projector lenses onto an imaging device (a fluorescent screen or a camera). As the electron beam has different degrees of transmission through different parts of the sample, the details of the sample can be displayed by the imaging device.

Let us take a closer look at the electron gun and the magnetic lenses.

a Electron gun

Its function is to emit electrons and accelerate them to high speeds (Fig 3.2k). The cathode is heated up by an a.c. and emits electrons. These electrons will then be accelerated by an electric field produced by a high voltage V (40–400 kV) across the cathode and the anode.

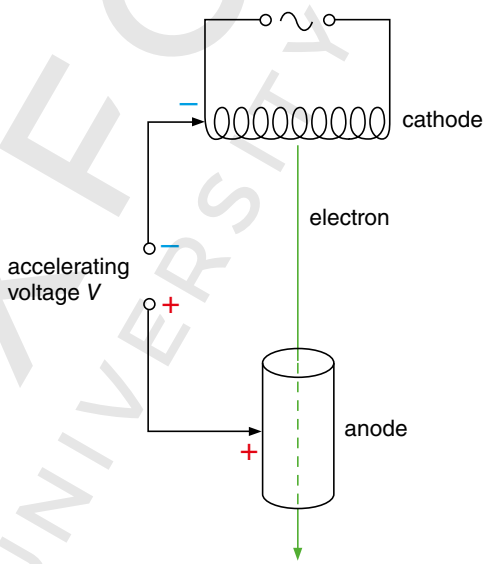


Fig 3.2k *The structure of the electron gun.*

For an electron with mass m and charge $-e$,

gain in KE = loss in electric PE

$$\frac{1}{2}mv^2 = eV$$

$$(mv)^2 = 2meV$$

Momentum of electron $p = mv = \sqrt{2meV}$

De Broglie wavelength of electron $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}} \propto \frac{1}{\sqrt{V}}$

Therefore, when the accelerating voltage V of the electron gun **increases**, the de Broglie wavelength λ of the electron **decreases**, and the **resolving power** of the TEM **increases**.

Example 5 | Resolving power of TEM

In a TEM, an accelerating voltage of 40 kV is applied across the cathode and the anode in the electron gun. The initial KE of electrons emitted from the cathode is negligible. Take the charge and mass of an electron as -1.60×10^{-19} C and 9.11×10^{-31} kg respectively, and Planck constant as 6.63×10^{-34} J s.

(a) When the electron leaves the electron gun, find its speed v and de Broglie wavelength λ.

(b) Estimate the angular resolution θ_R of the TEM by considering the Rayleigh criterion. Take the diameter of the aperture as 10 cm.

Solution

(a) Gain in KE = loss in electric PE

$$\frac{1}{2}mv^2 = eV$$

$$\therefore v = \sqrt{\frac{2eV}{m}} = \sqrt{\frac{2(1.60 \times 10^{-19})(40 \times 10^3)}{9.11 \times 10^{-31}}} = 1.185 \times 10^8 \text{ m s}^{-1}$$

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{(9.11 \times 10^{-31})(1.185 \times 10^8)} = 6.14 \times 10^{-12} \text{ m}$$

This is much shorter than the wavelength of visible light (~10^{-7} m).

(b) $\theta_R \approx \dfrac{1.22\lambda}{d} = \dfrac{1.22(6.14 \times 10^{-12})}{0.1} = 7.49 \times 10^{-11}$ rad

θ_R of an optical microscope is about 10^{-4} rad.

The resolving power of a TEM is also limited by other factors, such as the aberrations of the magnetic lenses.

Writing practice

Question

How will the resolving power of the TEM change if the accelerating voltage in the electron gun increases? (2 marks)

Answer

If the voltage increases, the electrons will move at a higher speed v. According to $\lambda = \dfrac{h}{mv}$, the de Broglie wavelength λ will decrease. 1A

According to $\theta_R \approx \dfrac{1.22\lambda}{d}$, θ_R will decrease, i.e. the resolving power will increase. 1A

Practise yourself

Why does a TEM has a higher resolving power than an optical microscope? (2 marks)

b Magnetic lenses

The electron beam has to pass through several magnetic lenses before it forms an image of the sample. These lenses are copper coils which produce strong magnetic fields to deflect the electron beam. Their function is similar to the convex lenses in an optical microscope (Fig 3.2l). The size of the current through the coil determines the focal length of a magnetic lens and, in turn, the magnification of a TEM.

The electrons should have the same speed. Otherwise, they will be deflected by different amounts by the magnetic lenses and the image will be blurred. ▶

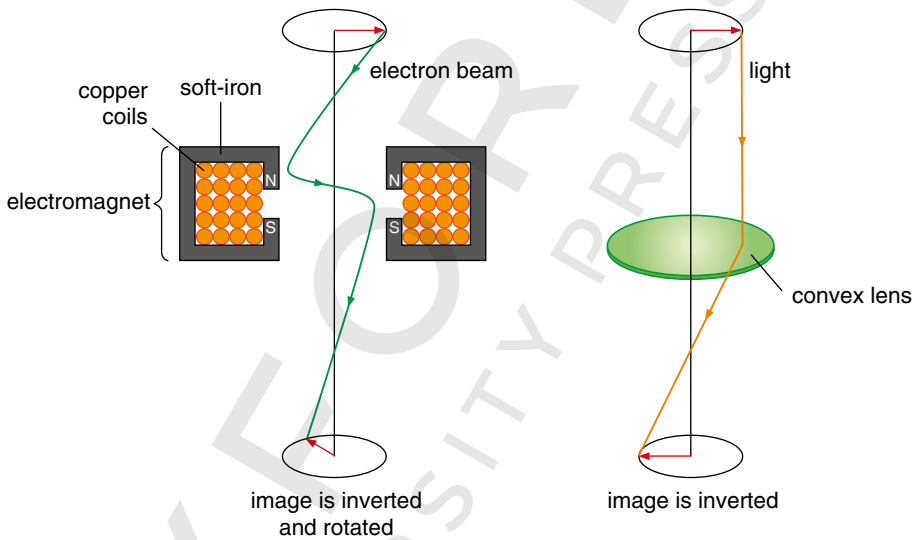


Fig 3.2l *Comparing the action of a magnetic lens with an optical lens.*

In a TEM, fast-moving electrons are directed through a thin sample to form an image.

The magnetic lenses produce magnetic fields to converge the electron beam, like a convex lens converging light.

Increasing the accelerating voltage of the electron gun will increase the speeds of the electrons and reduce their de Broglie wavelengths, so the TEM will attain a higher resolving power.

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) A TEM has a higher resolving power than an optical microscope because the electrons used by a TEM has a longer wavelength than visible light. (T/F)

(b) The resolving power of a TEM is higher when the accelerating voltage of its electron gun is higher. (T/F)

(c) The magnetic lenses in a TEM produce magnetic fields to converge the electron beam. (T/F)

☆ HKDSE 2014 Paper 2 Q.2(a)

2 Arrange the parts in a TEM in the order that an electron beam passes through.

(1) condenser lens, (2) objective lens, (3) projector lens, (4) viewing screen, (5) cathode, (6) anode, (7) sample

Learn by practice 3

Take $m_e = 9.11 \times 10^{-31}$ kg, $e = 1.60 \times 10^{-19}$ C and $h = 6.63 \times 10^{-34}$ J s.

1 In a TEM, the accelerating voltage of the electron gun is set at 80 kV. The initial KE of electrons emitted from the cathode is negligible.

(a) What is the speed of an electron emitted from the electron gun?

(b) What is the de Broglie wavelength of the electron in (a)?

(c) If the aperture of the TEM has a diameter of 10 cm, estimate the angular resolution of the TEM according to Rayleigh criterion.

2 The electron gun in a TEM emits electrons with a de Broglie wavelength of 0.02 nm. The initial KE of electrons emitted from the cathode is negligible.

(a) Find the speed of the electrons.

(b) Find the accelerating voltage of the electron gun.

(c) The accelerating voltage increases.

(i) How will the de Broglie wavelength of the electrons change?

(ii) How will the resolving power of the TEM change?

☆ HKDSE 2013 Paper 2 Q2.7, 2020 Paper 2 Q2.6

→ This videos demonstrates how this question is solved.

DSE practice 2 | Change in resolving power

Video

The minimum length that a TEM can resolve is 0.5 nm when the accelerating voltage is V. Microscope X uses protons instead of electrons to form images. The settings, such as the aperture diameter, the distance between the aperture and the sample, and the size of the accelerating voltage, are the same for both microscopes. The magnitude of the charge of a proton is the same as that of an electron, but its mass is 1800 times that of an electron. Estimate the minimum length that microscope X can resolve.

A 2.8×10^{-4} nm

B 0.012 nm

C 21 nm

D 900 nm

Solution

By $s \approx r\theta_R \approx r \times \dfrac{1.22\lambda}{d}$ and $\lambda = \dfrac{h}{\sqrt{2meV}}$,

See p.118.

$s \approx r \times \dfrac{1.22h}{d\sqrt{2meV}} \propto \dfrac{1}{\sqrt{m}}$ (for constant r, d and V)

Common mistake ⚠
Students may not be able to find this relationship.

$\dfrac{s_p}{s_e} \approx \dfrac{\sqrt{m_e}}{\sqrt{m_p}}$

$s_p \approx \dfrac{\sqrt{m_e}}{\sqrt{m_p}} \times s_e = \dfrac{\sqrt{1}}{\sqrt{1800}} \times 0.5 = 0.012$ nm

∴ The answer is B.

Revision 3 Q10 (p.130)

DSE goal

Describe how an STM works (principles of the tunnelling effect are not required).

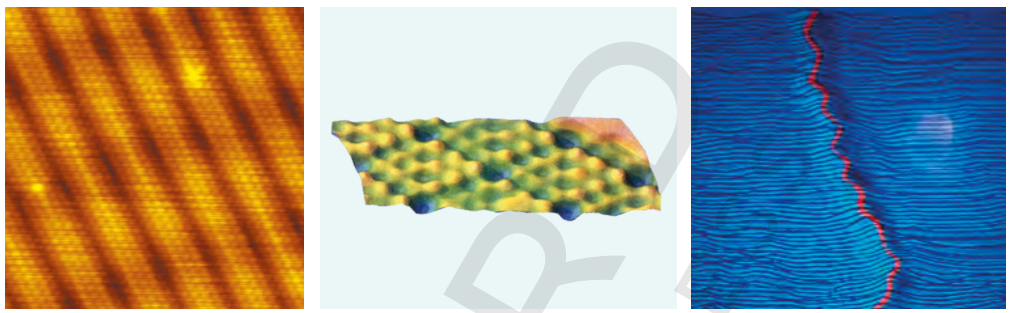
DSE exam

14(2)Q.2(c), 17(2)Q2.8, 18(2)Q2.7, 19(2)Q2.4, 20(2)Q2.7, 22(2)Q2.7

→ An introduction to quantum tunnelling can be found in the following website.

4 Scanning tunnelling microscopes

A **scanning tunnelling microscope** (STM) uses a different principle to attain atomic resolution. Figure 3.2m shows some images taken by STMs.



(i) Gold surface. *(ii) Silicon surface.* *(iii) DNA molecule.*

Fig 3.2m *STM images of some objects.*

An STM reveals the arrangement of atoms at a **surface** by using a fine conducting probe (Fig 3.2n) that moves over the surface. The tip of the probe, being formed by a single atom, is held extremely close to the sample at a distance of about an atom's diameter.



Fig 3.2n *The fine conducting probe in an STM.*

When a small potential difference is applied between the probe and the sample, some electrons jump between the sample surface and the probe. This phenomenon is called **quantum tunnelling**. The flow of electron produces an electric current, which is called the **tunnelling current** (Fig 3.2o). The tunnelling current increases drastically with a small decrease in the tip-to-surface distance.

▶ The principles of quantum tunnelling are out of the scope of this book.

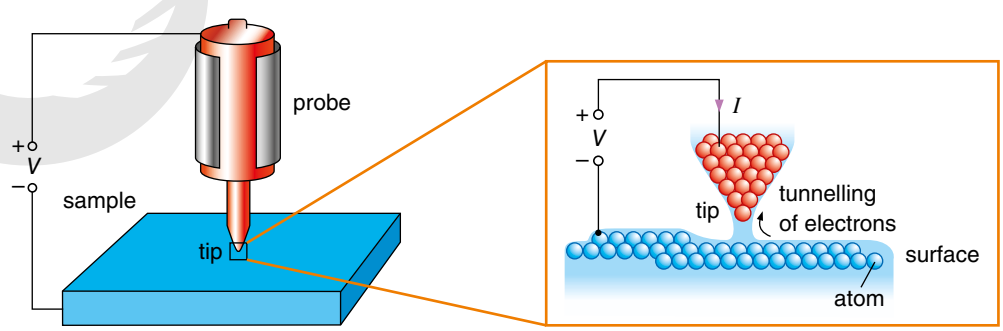


Fig 3.2o *Tunnelling current is produced between the sample surface and the tip.*

scanning tunnelling microscope 掃描隧穿顯微鏡　　quantum tunnelling 量子隧穿效應
tunnelling current 隧穿電流

The probe is controlled with precision to scan across the sample surface. Data is recorded and processed to produce the image. There are two methods of scanning (Table 3.2a).

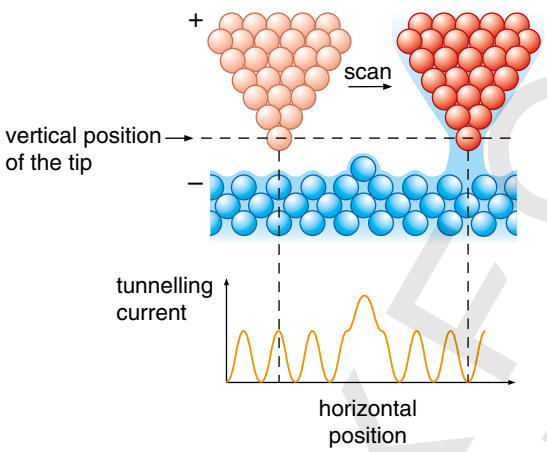
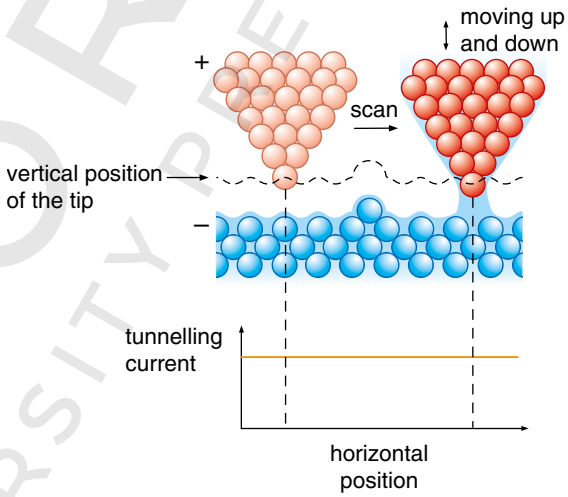
Scanning with constant height	Scanning with constant current
The probe scans horizontally across the sample surface at a **constant height**. The details of the surface are revealed by monitoring the **change in the tunnelling current**.	The probe scans across the sample surface with a **constant current**. The probe is moved up and down during scanning to keep the tip-to-surface distance unchanged. By monitoring the **vertical movement of the probe**, the details of the surface are revealed.
	

Table 3.2a *Scanning with constant height or constant current.*

The constant height method can only be applied to very flat surfaces (in atomic scale) as otherwise the tip of the probe may crash. On the other hand, the constant current method takes much longer time than the constant height method to finish the scan, as it requires precise adjustments to the vertical position of the probe.

Simulation

→ This simulation shows how a STM works. It is available on OUP web in PC version (.exe) only.

Figure 3.2p shows the basic set-up of an STM.

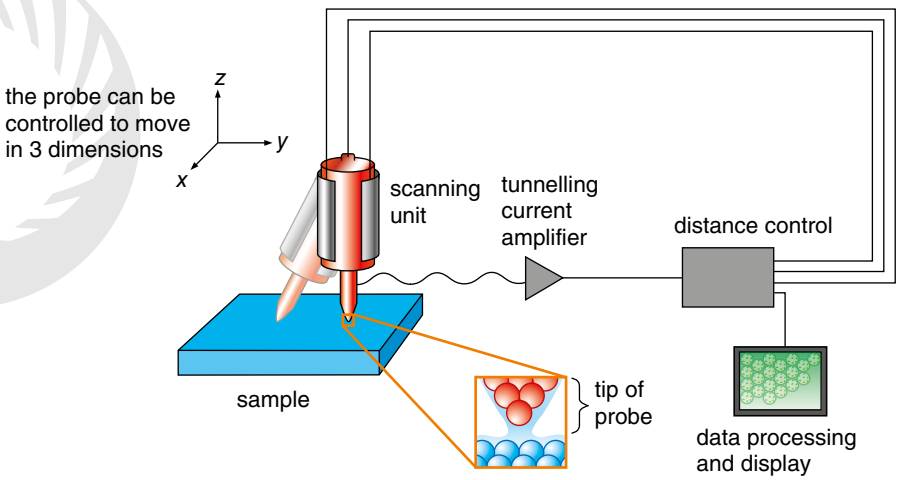


Fig 3.2p *Basic set-up of an STM.*

The resolving power of an STM is limited by the size of the probe's tip. The smaller the tip, the higher the resolving power. Since diffraction is not involved in the formation of the image, the Rayleigh criterion is **not** applicable to an STM.

A scanning tunnelling microscope makes use of the tunnelling current to produce an image of a sample surface with details down to atomic scale. It uses a probe to scan across the sample surface either at a constant height or with a constant current.

An STM can also be used to *manipulate* individual atoms. This allows an STM to extend beyond its probing function as a microscope to an instrument for building simple nano structures (Fig 3.2q).

This is the smallest movie ever created (frame size of 45 nm × 25 nm). Individual atoms were manipulated on a surface with an STM to produce 242 still images, which were then displayed in series to form the movie.

Video

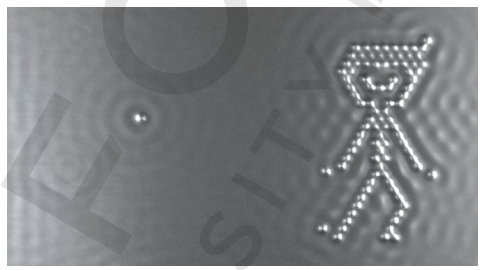


Fig 3.2q *'A Boy and His Atom', a movie created by using an STM.*

5 Comparing TEM and STM

Both a TEM and an STM can attain atomic resolution (~0.1 nm), but they work differently (Table 3.2b).

TEM	STM
use fast-moving electrons	use tunnelling current
resolution limited by de Broglie wavelength of electrons and size of aperture (Rayleigh criterion)	resolution limited by size of scanning tip
observe the inside of the sample	observe the surface of the sample
sample must be very thin	sample must conduct electricity
require ultra-high vacuum for operation to reduces the chance that the electron beam hits air molecules and gets deflected	does not require a vacuum, but works best in a high vacuum to avoid contamination of the sample
produce 2D images	produce 2D and 3D images
cannot move atoms	can move atoms

Table 3.2b *Comparison of TEM and STM.*

manipulate 操縱

Quick check 6

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 An STM can only work on a (thin / thick / conducting / non-conducting) sample. It shows the (surface / internal) structure of the sample.

2 In an STM, as the tip of the probe moves over the surface being observed at an extremely close distance, a tunnelling current is produced. This current (increases / decreases) as the tip-to-surface distance increases.

3 *True or false:*

(a) An STM can be used to move atoms. (T/F)

(b) The resolving power of an STM is limited by diffraction. (T/F)

(c) The resolving power of an STM can be increased by increasing the potential difference between the sample and the tip. (T/F)

4 Determine whether the following statements describe (A) a TEM or (B) an STM.

(a) The sample must be very thin.

(b) It observes the surface of the sample.

(c) It can manipulate atoms.

(d) It can form images of a plastic sample.

(e) Its resolving power can be varied easily.

Exercise 3.2

Take $m_e = 9.11 \times 10^{-31}$ kg, $e = 1.60 \times 10^{-19}$ C and $h = 6.63 \times 10^{-34}$ J s.

1 Which of the following is the phenomenon that a scanning tunnelling microscope mainly makes use of in imaging?

A Heating effect of current

B Photoelectric effect

C Tunnelling effect of electrons

D Electromagnetic induction

2 Which of the following components is/are used in a transmission electron microscope?

(1) Electron gun

(2) Light source

(3) Fine conducting probe

A (1) only

B (3) only

C (1) and (2) only

D (1) and (3) only

★**3** Which of the following statements about an optical microscope and a transmission electron microscope is/are correct?

(1) The resolving power of both microscopes are affected by the size of the aperture.

(2) Optical lenses are used in both microscopes to form images.

(3) Diffraction accounts for the limited resolving power of both microscopes.

A (1) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★**4** Which of the following statements about a scanning tunnelling microscope and a transmission electron microscope is/are correct?

(1) Both of them use fast-moving electrons to hit the sample.

(2) Both of them make use of tunnelling effect of electrons.

(3) Both of them can attain atomic resolution.

A (3) only

B (1) and (2) only

C (1) and (3) only

D (1), (2) and (3)

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*5 Which of the following changes will increase the minimum angular separation that can be resolved by a TEM?

(1) Reduce the acceleration voltage of the electron gun.

(2) Increase the aperture of the magnetic objective lens.

(3) Decrease the distance between the sample and the magnetic objective lens.

A (1) only

B (2) only

C (1) and (3) only

D (1), (2) and (3)

*6 The tip of the probe in a scanning tunnelling microscope scans horizontally with a constant height over a surface as shown in Figure a.

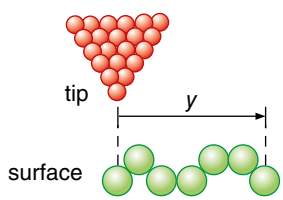
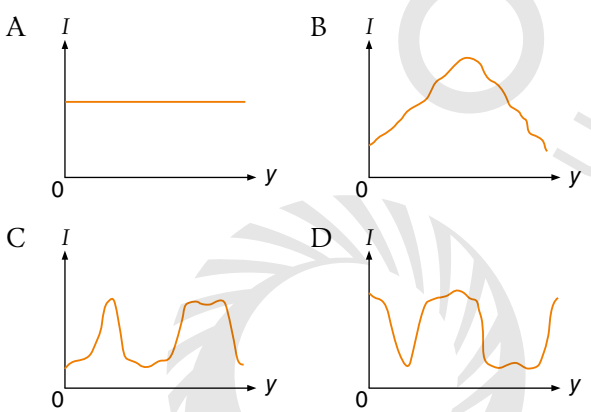


Fig a

Which of the following graphs correctly shows the variation of the tunnelling current I with the horizontal distance y travelled by the tip?



*7 The accelerating voltage in a transmission electron microscope is set to 50 kV. The initial KE of the electrons emitted from the cathode is negligible.

(a) Find the de Broglie wavelength of an electron accelerated by this voltage.

(b) How will the angular resolution of the microscope change when the accelerating voltage increases? Explain.

*8 What is the order of magnitude of the minimum angular separation between two points which can be resolved by

(a) the human eye (pupil diameter 4 mm) under normal lighting,

(b) a radio telescope (effective aperture 300 m) observing EM waves of frequency 10^8 Hz?

*9 Tom uses an optical microscope to examine a blood sample. He places the sample at the focus of the objective (Fig b). The diameter of the objective is the same as its focal length f. Take the wavelength of the light as 680 nm.

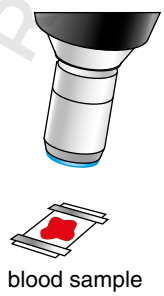


Fig b

(a) Estimate the minimum angular separation θ_R that the microscope can resolve by considering the Rayleigh criterion. Express your answer in terms of f.

(b) Estimate the minimum distance between two points in the sample that can be resolved.

(c) Explain why a TEM has a higher resolving power than an optical microscope.

*10 The screen of a mobile phone has dimensions of 11.52 cm × 6.48 cm with 3840 × 2160 pixels.

(a) What is the distance between two adjacent pixels?

(b) David is using the phone at a distance of 40 cm from his eyes (Fig c). The pupil of David's eye has a diameter of 5 mm. The wavelength of visible light is 400–700 nm. Is he able to distinguish the individual pixels? Show with calculation.

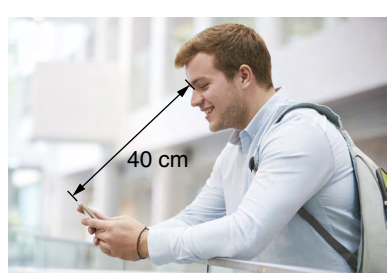
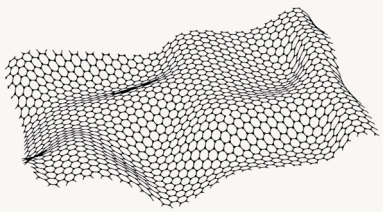
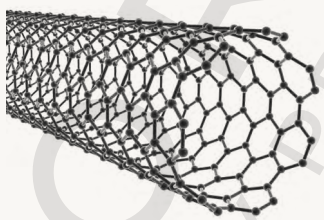
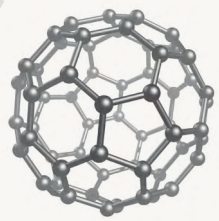


Fig c

Key concepts 3

1 Nano material

- Materials with at least one dimension in the range 1 nm to 100 nm
- Examples of nano structures of carbon:

	graphene	carbon nanotube	buckyball
			
Dimensions in nano scale	1	2	3

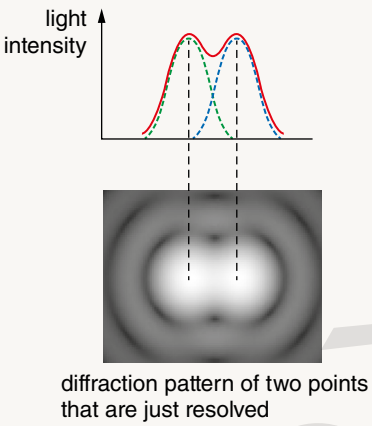
- Physical properties that may be different from bulk form:
  - Electrical property: conductivity
  - Mechanical property: strength and hardness
  - Optical property: colour and transparency
- Reasons of having different properties:
  - Different arrangement of atoms
  - Surface area effect: nano form has a higher surface-area-to-volume ratio
  - Quantum effect
- Applications:
  - Detoxicating, deodorizing and antimicrobial materials
  - Chemical and biological sensors
  - Water-repelling, stain-resistant and self-cleaning materials
  - Lighter and stronger equipment
  - Processors in electronic equipment
- Potential hazards:
  - Nano materials may be harmful to public health and the environment

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2 Rayleigh criterion

Resolving power limited by diffraction

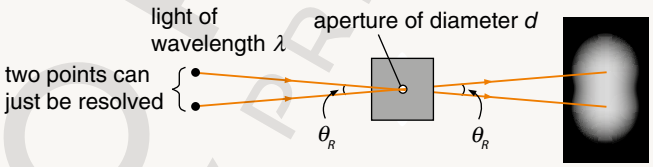
- Diffraction occurs when light passes through the aperture of an instrument.
- Two points can just be resolved when the central maximum of the diffraction pattern of one point is centred on the first-order minimum of the diffraction pattern of the other.



- Minimum angular separation between two points which can be resolved:

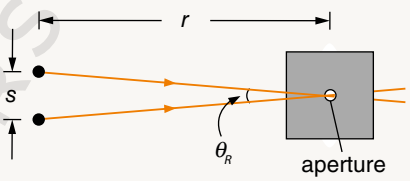
$$\theta_R \approx \frac{1.22\lambda}{d}$$

The smaller the value of θ_R, the higher the resolving power.



- The minimum distance s between two points that can be resolved:

$$s \approx r\theta_R$$



3 Microscopes for seeing in nano scale

Transmission electron microscope (TEM)

- Use fast-moving electrons to transmit through the sample
- Sample must be very thin
- Accelerate electrons with an electron gun
- Use magnetic lenses
- High accelerating voltage V

  $\Rightarrow$ high speed of electrons ($v = \sqrt{\frac{2eV}{m_e}}$)

  $\Rightarrow$ short de Broglie wavelength ($\lambda = \frac{h}{mv}$)

  $\Rightarrow$ high resolving power ($\theta_R \approx \frac{1.22\lambda}{d}$)

Scanning tunnelling microscope (STM)

- Make use of tunnelling current between the tip and the sample surface, which increases as the separation decreases
- Sample must be electrically conducting
- Have a fine conducting probe with its tip formed by a single atom
- Resolving power limited by the size of the probe's tip
- Scan across the sample surface by two methods:
  - constant height
  - constant current

Key terms

1 angular resolution 角解像度 p.112
2 angular separation 角間距 p.112
3 Lotus effect 蓮花效應 p.104
4 nanometre 納米 p.96
5 quantum tunnelling 量子隧穿效應 p.122
6 Rayleigh criterion 瑞利判據 p.111
7 resolution 解像度 p.110
8 resolving power 分辨率 p.110
9 scanning tunnelling microscope 掃描隧穿顯微鏡 p.122
10 transmission electron microscope 透射電子顯微鏡 p.117
11 tunnelling current 隧穿電流 p.122

Revision 3

Take $m_e = 9.11 \times 10^{-31}$ kg, $e = 1.60 \times 10^{-19}$ C and $h = 6.63 \times 10^{-34}$ J s.

Multiple-choice questions

1 3.2 What are the 'lenses' used in a transmission electron microscope made of?

A Glass
B Diamond
C Electromagnets
D Electrons

2 3.1 Which of the following properties does nano paint containing TiO_2 nanoparticles have?

(1) Antibacterial
(2) Blocking ultra-violet radiation
(3) Self-cleaning

A (1) and (2) only B (1) and (3) only
C (2) and (3) only D (1), (2) and (3)

***3** 3.2 Which of the following are the advantages of an STM over a TEM?

(1) An STM has a much higher resolving power.
(2) An STM can manipulate atoms.
(3) It is not necessary to make the sample very thin in using an STM.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

***4** 3.2 The minimum angular separation between two point sources of violet light which can be resolved by John's eyes is θ. Which of the following could be the angular separation between two point sources of red light which can be resolved by his eyes?

(1) 0.6θ
(2) 1.5θ
(3) 2θ

A (3) only B (1) and (2) only
C (2) and (3) only D (1), (2) and (3)

***5** 3.2 The James Webb Space Telescope (JWST) has a higher resolving power than the Hubble Space Telescope (HST) (Fig a). The diameter of the aperture of JWST is 6.5 m while that of HST is 2.4 m. When observing in the same wavelength, if the angular resolution of HST is θ, estimate the angular resolution of JWST.

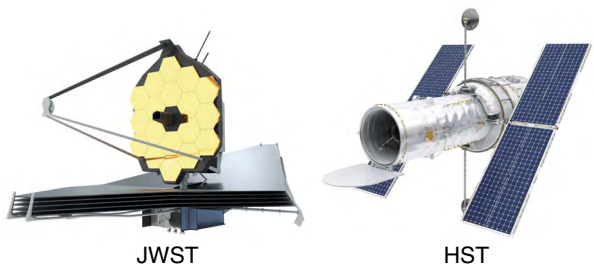


Fig a

A 0.14θ
B 0.37θ
C 2.7θ
D 7.3θ

***6** 3.1 Which of the following physical properties may change when a material is reduced to nano size?

(1) Melting point

(2) Transparency

(3) Electrical conductivity

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

***7** 3.2 Optical microscopes with apertures of different sizes and light of different frequencies are used to image samples. Which of the following combinations provides the highest resolving power?

	Size of aperture	Frequency of light
A	$2d$	f
B	$2d$	$2f$
C	d	f
D	d	$2f$

****8** 3.2 The angular resolution of a TEM is θ when the accelerating voltage of the electron gun is V. If the accelerating voltage is increased to $2V$, what is the angular resolution of the TEM?

A $\frac{1}{2}\theta$ B $\frac{1}{\sqrt{2}}\theta$

C $\sqrt{2}\theta$ D 2θ

Refer p.119, 121

****9** 3.2 A transmission electron microscope uses electrons of momentum p to form images. The minimum angular separation that it can resolve is θ. Suppose another microscope use particles of the same charge but momentum $2p$ to form images. Estimate the minimum angular separation that it can resolve.

A $\frac{1}{2}\theta$ B $\frac{1}{\sqrt{2}}\theta$

C $\sqrt{2}\theta$ D 2θ

Refer p.119, 121

10 3.2 **HKDSE 2013 Paper 2 Q2.7**

The minimum resolvable length of a typical transmission electron microscope (TEM) is about 0.2 nm. If a particle has the same charge of an electron and its mass is four times that of an electron, and a beam of such particles is accelerated through the same voltage in a TEM, the minimum resolvable length becomes

A 0.05 nm.

B 0.1 nm. (22%)

C 0.4 nm.

D 0.8 nm.

11 3.2 **HKDSE 2015 Paper 2 Q2.7**

Which of the following can increase the resolving power of a transmission electron microscope (TEM)?

(1) Increasing the anode voltage in the electron gun

(2) Decreasing the aperture of the magnetic objective lens

(3) Increasing the separation between the magnetic projection lens and the fluorescent screen

A (1) only (42%)

B (2) only

C (1) and (3) only

D (2) and (3) only

12 3.1 **HKDSE 2016 Paper 2 Q2.8**

Which of the following applications in nanotechnology utilize(s) **Lotus effect**?

(1) Water-repelling fabric used in swimming suits is manufactured by nano-coating.

(2) Glass is made self-cleaning by coating it with a water-attracting material in nanoscale.

(3) Nano-sized zinc oxide is added to fabric as a photocatalyst for protection from dirt.

A (1) only (32%)

B (1) and (2) only

C (1) and (3) only

D (2) and (3) only

13 3.1 **HKDSE 2018 Paper 2 Q2.8**

Which statement below is **incorrect**?

A Nano particles of silver show a colour different from the colour of silver in bulk form.

B Diamond is a poor thermal conductor and a poor electrical conductor as it does not have free electrons. (26%)

C A carbon nanotube is much stronger than steel of the same size.

D Carbon buckyballs such as C_{60} are formed by carbon atoms arranged in hollow cages.

14 3.2 **HKDSE 2019 Paper 2 Q2.7**

Two point sources of red light at a distance of 160 m from an observer can just be resolved by the naked eyes. If they are replaced by point sources of violet light, how should the observer move from the original position such that the two sources can just be resolved?

A Move about 280 m further away from the sources

B Move about 120 m further away from the sources (38%)

C Move about 120 m towards the sources

D Move about 70 m towards the sources

Conventional questions

★**15** 3.2 Figure b shows a typical optical microscope.



Fig b

(a) Which type of electromagnetic waves is used for imaging in an optical microscope? (1 mark)

(b) Which wave phenomenon limits the resolving power of an optical microscope? (1 mark)

(c) Take the diameter of the aperture of the microscope as 1 cm and the separation between the sample and the aperture as 2 cm. Show that the microscope cannot be used to observe nano materials. (3 marks)

(d) Explain how a transmission electron microscope can attain a much higher resolving power than an optical microscope. (2 marks)

★**16** 3.1 Silver vessels have long been used to store water and liquids (Fig c). Ancient people believed that silver vessels could keep the liquids 'clean'.



Fig c

(a) State one property of silver that makes the liquids 'clean'. (1 mark)

(b) Silver nanoparticles are mixed with other materials to make a container. What is the advantage of this method compared to using silver in bulk scale to make the whole container? Explain briefly. (2 marks)

(c) Find the surface-area-to-volume ratios of a silver cube of side length

(i) 1 cm, (1 mark)

(ii) 80 nm. (1 mark)

(d) State another daily-life application of nanotechnology. (1 mark)

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★**17** 3.1 Carbon atoms can be arranged in a long cylindrical tube (Fig d). These tubes are known as carbon nanotubes (CNTs).

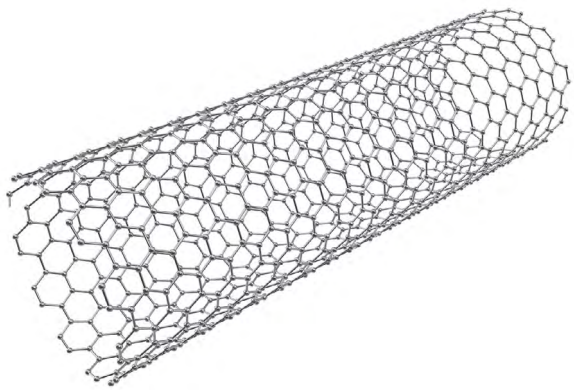


Fig d

(a) What does 'nano' mean? (1 mark)

(b) CNTs are 100 times stronger than steel, but only $\frac{1}{6}$ as heavy. Give one application of CNTs that makes use of these properties. (1 mark)

(c) State two other forms of carbon in nano scale. (2 marks)

(d) Scientists are worried about the spread of nano materials in the environment. Give two reasons why they pay more attention on the health risks caused by the same materials in nano scale than in bulk scale. (2 marks)

★**18** 3.1 Both the normal-sized and nano-sized zinc oxide (ZnO) particles can be used in sunscreens.

(a) Why is ZnO used in sunscreens? (1 mark)

(b) What is the advantage of nano-sized ZnO particles over normal-sized ZnO particles when used in sunscreens? (1 mark)

(c) In some countries, the use of sunscreens with nano-sized ZnO particles is prohibited. Give one possible reason for this. (1 mark)

(d) State another application of nanotechnology. (1 mark)

★**19** 3.2 Figure e shows a scanning tunnelling microscope (STM).

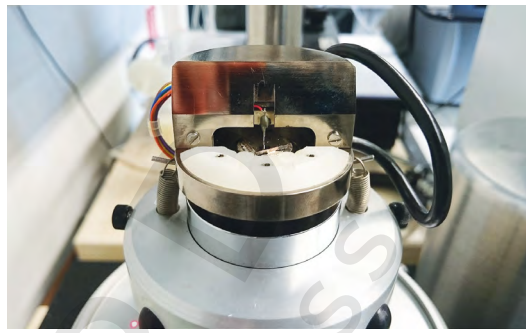


Fig e

(a) Explain how an STM can attain atomic resolution. (3 marks)

(b) State a requirement for a sample to be imaged by an STM. (1 mark)

(c) A transmission electron microscope (TEM) can also attain atomic resolution. Suggest a situation in which

(i) an STM is preferred over a TEM,

(ii) a TEM is preferred over an STM

in studying a sample. Assume the sample fulfils the requirement in (b). (2 marks)

★★**20** 3.2 A transmission electron microscope (TEM) is able to resolve details down to nano scale.

(a) Express 1 nm in metres. (1 mark)

(b) In a TEM, the accelerating voltage of the electron gun is set to 60 kV.

(i) State the function of an electron gun. (1 mark)

(ii) Estimate the wavelength of the electrons after passing the anode of the electron gun. The initial KE of electrons emitted from the cathode is negligible. (3 marks)

(iii) The accelerating voltage is then adjusted to make the wavelength of the electrons longer. How does the resolving power of the TEM change? Explain briefly. (2 marks)

(c) (i) Briefly describe the function of the electromagnetic lenses in a TEM. (1 mark)

(ii) Hence, explain why the electrons emitted from the electron gun should have the same speed. (2 marks)

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(d) State a requirement of the sample imaged by a TEM. (1 mark)

Refer p.118–120

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3.2

(a) In a transmission electron microscope (TEM), electrons emitted from the cathode pass through the specimen and the four functional parts listed below before forming an image on a screen.

Functional parts:

(1) objective magnetic lens

(2) projection magnetic lens

(3) condensing magnetic lens

(4) anode

Referring to the following block diagram of a TEM, match the functional parts represented by *A*, *B*, *C* and *D* in the diagram (Fig f). (2 marks)

cathode	*A*	*B*	specimen	*C*	*D*	screen

Fig f

(b) (i) When an electron of mass m and charge e is accelerated from rest by a voltage V, show that its de Broglie wavelength λ is given by $\lambda = \frac{h}{\sqrt{2meV}}$ where h is the Planck constant. (2 marks)

(ii) The accelerating voltage of a TEM is 10 kV. Find λ. (2 marks)

(iii) Explain why the resolving power of a TEM is higher compared with an optical microscope. (2 marks)

(c) Both scanning tunnelling microscopes (STM) and transmission electron microscopes (TEM) have very high resolving powers. Now if the internal structure of a slice of metallic specimen is to be studied, which of the above microscopes would be suitable or are both suitable? Explain. (2 marks)

Q21 Exam Report: Part (a) revealed that many candidates were not familiar with the structure of TEM. Candidates performed well in (b). In (b)(i), most were able to derive the formula, however, some of them did not state conservation of energy explicitly. Most candidates substituted correct values into the formula to obtain the answer in (b)(ii) except for some careless mistakes, such as wrong or missing units. In (b)(iii), many candidates knew that the wavelength of an electron is smaller than that of visible light. Weaker ones misunderstood that a larger value of θ implied higher resolving power. In fact θ is the angular separation between objects such that a smaller value means that it is possible to distinguish objects of smaller separation, and thus a higher resolving power. Candidates' performance in (c) was poor. Many mixed up the features and principles of TEM and STM.

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Nature of science

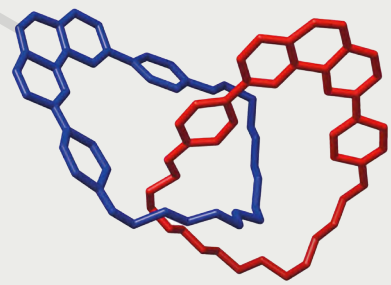
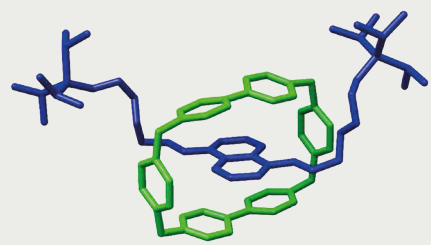
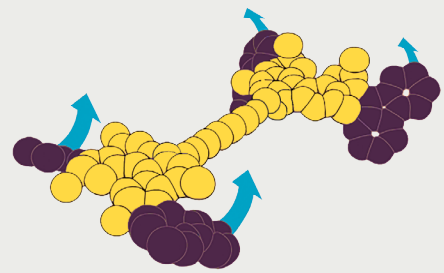
Difference between science and technology

Science is different from technology, though they are closely related. 'Science' comes from the Latin word 'scientia' which means knowledge. We attempt to understand the natural world through scientific investigation. 'Technology' comes from the Greek word 'techne' which means *artefact*, art or *craft*. It can be a physical object, e.g. a smartphone; or a technique or skill, e.g. the technology of making stained glass. Technology is often driven by demands of society, which may exist because of good or selfish intentions. The technology may come before or after the science behind. **Nanoscience** is concerned with **understanding the properties** of materials in nano scale, while **nanotechnology** aims to **make use of the properties to create** novel structures, skills or devices for a wide range of applications.

Know more

Molecular machines (p.104)

In 2016, Jean-Pierre Sauvage, Fraser Stoddart and Ben Feringa were awarded the Nobel Prize in Chemistry for their pioneering works in developing molecular machines. Their works are shown below.

 *Jean-Pierre Sauvage (1944–)*	In 1983, Sauvage and his group developed a way to link two ring-shaped molecules to form a chain. Later, they developed a way to control the ring's rotation.	
 *Fraser Stoddart (1942–)*	In 1991, Stoddart and his group developed a method to thread a molecular ring onto another thin molecular track and control the movement of the ring along the track.	
 *Ben Feringa (1951–)*	In 1999, Feringa developed a molecular motor that spun continuously in the same direction. His team then constructed a four wheel molecular car by making a molecular motor at each of the four wheels.	

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